



# Enhancing Systems Engineering Academic Processes with AI at NPS: Initial Experiences

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# Acknowledgments

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# Introduction

- Systems Engineering graduate education depends on rigorous but labor-intensive academic processes.
- Education technician support, advising, thesis/capstone assessment, grading, and accreditation evidence require consistency and traceability.
- Staff and faculty time are constrained, while institutional and defense-relevance expectations remain high.
- The SE Department is using governed AI assistants as early adopters for practical campus implementation.
- The goal is not autonomous decision making, but better human-reviewed academic products, evidence, feedback, and process continuity.

## Current AI Assistant Portfolio

- **Higher Education Administrative Tool Helping Engineers Renew (HEATHER 2):** supports education technician and academic administration processes using SOPs and policy documents.
- **Capstone/Thesis Reviewer:** generates rubric-based reviews and structured assessment outputs for faculty review and student feedback.
- **Automatic Graders:** AI assisted workflows for course-level grading of assignments and formative feedback.
- These efforts are being executed locally in SE while establishing reusable patterns for broader NPS adoption.

## Current AI Impediments

- Non-governmental LLMs requiring personal accounts.
- Privacy restrictions using non-governmental LLMs hamper capabilities.
- Lack of government resources to fully support these applications.
- Problem domain difficulties and LLM performance.
- Cloud computing costs.
- Local machine LLMs can alleviate some of the above.

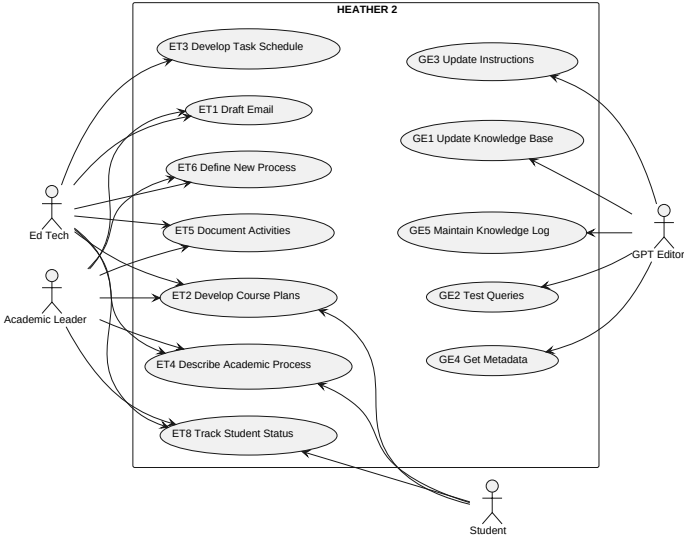
## HEATHER 2: Education Technician Support

- Custom GPT grounded in SE Department Standard Operating Procedures (SOPs).
- Supports matriculation, ABET checks, enrollment management, student matrices, graduation, and routine communications.
- In routine daily usage by Ed Tech and faculty.



*Illustration by HEATHER 2*

# Representative HEATHER 2 Use Cases



# Current Documents in HEATHER 2 Knowledge Base

| Document                                               | Summary                                                                                                                                               |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Business Continuity Plan for SE (DL) Operations        | Procedures to sustain DL academic operations during outages or disruptions.                                                                           |
| ABET SOP for MSSE Program                              | Workflow for ABET/equivalency verification to determine MSSE degree eligibility.                                                                      |
| Graduation Process SOP                                 | Timelines, forms, and approval chains for SE DL degree and certificate completion.                                                                    |
| New Student Enrollment Cheat Sheets                    | Procedures for Python matrix assignment, track designation, and academic commenting.                                                                  |
| Matrix Management Cheat Sheets                         | Instructions for editing matrices, verifying certificate eligibility, and updating EDD/EGD.                                                           |
| Catalog Update Procedures                              | Step-by-step instructions for updating SmartCatalog entries for SE curricula.                                                                         |
| Applicant Reporting Cheat Sheets                       | Filters, date ranges, and formatting guidelines for generating applicant data from Python.                                                            |
| INCOSE Certification Support SOP                       | Eligibility criteria and communication workflow for ASEP/CSEP via SE3100 equivalency.                                                                 |
| Outstanding Thesis Selection SOP                       | End-to-end nomination, committee, and approval workflow for Outstanding Thesis awards.                                                                |
| Administrative and Academic Process Guides             | High-level guidance on recurring responsibilities, communications, and records practices.                                                             |
| Education Technician Position Description (GS-1702-09) | Federal classification guidance on core Ed Tech responsibilities, authority, and evaluation.                                                          |
| heather.py                                             | Current usage model for the GPT with defined use cases, associated testing data, and saved chat histories for self awareness, auditing and reporting. |



# Example Ed Tech Use Case Testing Results

| ID    | Use Case Name               | Tester(s) | Date       | Result | Test Session Links                               |
|-------|-----------------------------|-----------|------------|--------|--------------------------------------------------|
| ET1   | Draft Email                 | Madachy   | 2025-04-22 | Pass   | <a href="#">Student Email</a>                    |
| ET1   | Draft Email                 | Madachy   | 2025-04-22 | Pass   | <a href="#">Cohort Email</a>                     |
| ET1   | Draft Email                 | Samaha    | 2025-04-29 | Pass   | <a href="#">Draft Sponsor Emails</a>             |
| ET2   | Develop Course Plans        | Group     | 2025-04-25 | Pass   | <a href="#">Standard Course Plan</a>             |
| ET2   | Develop Course Plans        | Bell      | 2025-04-26 | Pass   | <a href="#">PhD Remediation Plan</a>             |
| ET2.1 | Develop Custom Course Plans | Group     | 2025-04-25 | Pass   | <a href="#">Customized Course Plan</a>           |
| ET3   | Develop Task Schedule       | Madachy   | 2025-04-22 | Pass   | <a href="#">Graduation Task Plan</a>             |
| ET3   | Develop Task Schedule       | Samaha    | 2025-05-02 | Pass   | <a href="#">Ed Tech Calendar Analysis</a>        |
| ET4   | Describe Academic Process   | Bell      | 2025-04-26 | Pass   | <a href="#">Candidacy Steps</a>                  |
| ET4   | Describe Academic Process   | Bell      | 2025-04-26 | Pass   | <a href="#">PhD Exams</a>                        |
| ET4   | Describe Academic Process   | Madachy   | 2025-04-22 | Pass   | <a href="#">Draw ABET Accreditation Sequence</a> |
| ET5   | Document Activities         | Samaha    | 2025-04-29 | Pass   | <a href="#">Progress Reporting</a>               |
| ET6   | Define New Process          | Madachy   | 2025-04-28 | Pass   | <a href="#">Document process improvement</a>     |

# HEATHER 2 Example: Evaluate Proposed Student Course Changes

Here is the review and approval assessment for each of the students listed under 311-AA Ray Madachy's purview, based on the provided course planning, matrix alignment, and degree requirements for MSSE and related certs (131, 133, 282), and compliance with 2 of 7 SOPs for ABET, graduation, and scheduling.

## ✓ 1. [REDACTED] (311-253C) – Israeli DL Student

Status: APPROVE

Assessment:

- His future course schedule fills out all remaining MSSE requirements.
- Fully integrated into existing 311-251G and 311-262X cohort groups.
- Courses spread across quarters to ensure continuity and cohort synergy.
- 131, 133, and 282 certs all captured with courses like SE3050, SE3011, SI3400, SE3302, SE4930, SE4150/4151, SE4354, SE4003, SE3201-3203.
- No overload or term conflicts.
- Non-ABET/ABET status not specified, but nothing precludes proceeding.
- Off quarter in 27/1 is acceptable for international students.

✓ Recommend moving forward with schedule.

## ✓ 2. [REDACTED] (282-244C)

Status: APPROVE

Assessment:

- In 282 cert only. Certificate students are not required to follow full MSSE progression.
- Adjustment accounts for canceled offerings (SI3400 and SE3302), with reassignment to 311-262X sections.
- Completes SI3400 and SE3302 with minimal delay; standard course pacing for a certificate student.
- No overload, no conflicts. Sequencing logical.

✓ Approve schedule update with replacement SI3400/SE3302 in 27/1 and 26/4.

## ✓ 3. [REDACTED] (311-252C) – Israeli DL Student

Status: APPROVE WITH NOTE

Assessment:

- Enrolled in 311 MSSE and meets degree pathway via proposed plan.
- Suggestion to drop SE3050 in 26/1 and replace with SI3400 directed study with Joe Sweeney is pedagogically sound and avoids overload. SI3400 asynchronous accommodates Israeli time zone.
- 131 and 133 cert requirements are covered.
- Multiple paths built into plan (SE4354 vs SE4950), allowing flexibility based on offerings.
- Course load per quarter remains manageable.

✓ Recommend approval with Joe Sweeney's concurrence on the directed study.

...

## ✓ 5. [REDACTED] (721-251C)

Status: APPROVE

Assessment:

- Dual certs and MSSEM pathway.
- Cert 133 courses (SE3900, SE4930, SE4935, SE4420) previously completed.
- Later moves into 171 and SE methods track for 721 MSSEM:
  - SE4354, SE3302, SE4520, SE4353
  - SI4021, SE3011 = solid SE methods
- Has 36 units of 4-unit courses and at least 4 at 4000-level.
- Needs 1 additional 4-unit class to reach 48 units — courses from former cert 133 count (per NPS policy).
- Thesis remains pending. Once started and completed, all degree requirements will be met.

✓ Recommend approval with confirmation that thesis enrollment (SE0811) is monitored and graded on time.

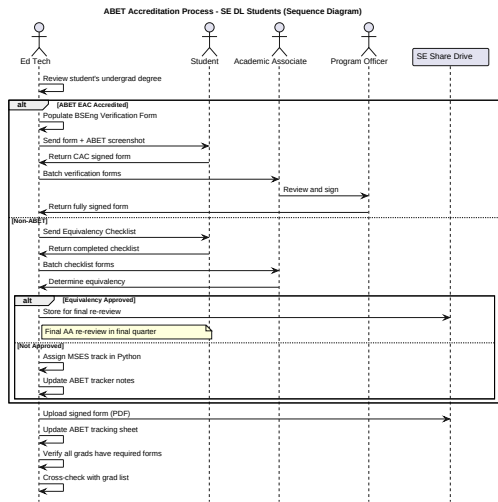
## Additional Action Items for Ed Tech:

- For [REDACTED] and [REDACTED], confirm ABET/EAC or equivalency status and ensure the appropriate form is filed per [ABET SOP].
- For all 311 students, confirm updated EGD/EDD in Python and annotate matrix comments as per SOP.
- Ensure directed study for SI3400 ([REDACTED]) is set up in Python with "DL" box checked and instructor approval documented.
- Lock matrix once confirmed.

## Summary:

| Student    | Curriculum | Status | Approval     | Notes                                             |
|------------|------------|--------|--------------|---------------------------------------------------|
| [REDACTED] | 311        | Active | ✓<br>APPROVE | Schedule fulfills full MSSE and cert requirements |
| [REDACTED] | 282        | Active | ✓<br>APPROVE | Course substitutions are valid                    |

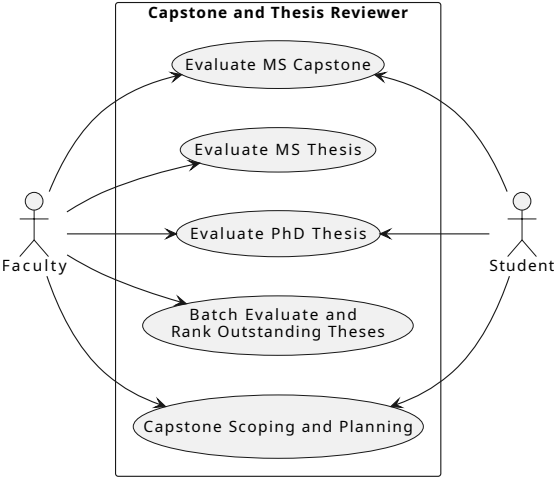
# HEATHER 2 Example: Describe Academic Process



## Capstone/Thesis Reviewer

- Faculty-facing assistant for assessment of draft or final theses and capstone reports.
- Applies SE rubrics, defense relevance, technical rigor, and writing criteria to assess NPS learning outcomes
- Produces coordinated narrative reviews and spreadsheet assessment workbooks.
- Current work is refining instructions, calibration, and experiments against faculty judgments and golden examples.
- Results support student feedback, accreditation evidence, and cohort-level analysis.
- Emergent use cases includes batch assessment for outstanding thesis ranking and student usage.

# Demonstrated Capstone/Thesis Reviewer Use Cases

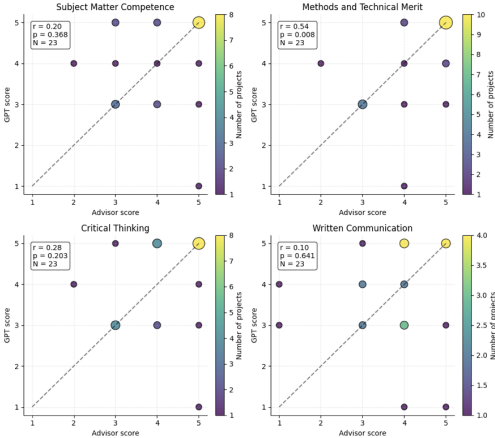


## Assessment Approach

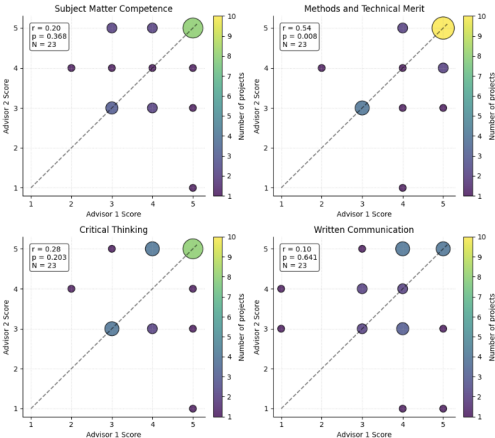
- Compare AI outputs with human baselines while recognizing variability among human reviewers.
- Use paired statistical tests, effect sizes, correlation analysis, and qualitative faculty review where appropriate.
- Clarify sample sizes and treat early statistical results as calibration evidence, not final proof.
- Calibrate against verified golden examples rather than optimizing only to historical advisor scores.
- Track review time, rework, consistency, completeness, student/faculty trust, and downstream assessment value.

# Interim Assessment Calibration Analysis

Advisor vs GPT Scores by Learning Outcome  
Bubble size and color = number of projects



Advisor 1 vs Advisor 2 Scores by Learning Outcome  
Bubble size and color = number of reports



# Example Capstone Review

## Capstone Review Summary

Student Names: [REDACTED]

Curriculum: MS Systems Engineering Management (curriculum code not specified in draft)

Review Date: 06 Feb 2026

## Thesis Office Format-Check Readiness (Gate Assessment)

Overall assessment: Not ready to submit for Thesis Processing Office (TPO) format checking.

### Format/administrative blockers that must be corrected first

- **⚠️** Front matter is explicitly marked as a "preview for proofreading ONLY," and instructs that corrections must be made in the thesis dashboard and that the preview pages will be replaced during final review. This indicates the file is not the official assembled output intended for processing. (See the initial preview pages and instructions.)  
→ Recommendation: Regenerate/export the official capstone PDF from the NPS thesis dashboard after front-matter fields are final, then use that output for format checking.
- **⚠️** Signature/approval blocks are incomplete: "Lead Editor:" and "Reviewed by:" are blank in the signature page area.  
→ Recommendation: Complete required fields in the dashboard so the exported PDF contains the correct official blocks.
- **⚠️** Sponsor field shows "N/A" on the Report Documentation Page while the Acknowledgments names AFRL as sponsor and cites other stakeholders.  
→ Recommendation: Ensure sponsor/monitoring agency entries are consistent across RDP, acknowledgments, and any sponsor letters required by process.

**Conclusion for TPO gate:** Resolve the above issues and regenerate the official PDF before requesting format checking.

## Technical Content and Systems Engineering Rigor (Advisor-Focused)

### Strengths (technical and contribution)

- **✅** Clear problem framing and defense-relevant motivation: the report describes the challenge of making risk acceptance and cybersecurity assurance decisions under cost/schedule constraints for complex microelectronics supply chains and integrated HW/SW systems.
- **✅** Coherent solution concept: the Assurance Bill of Materials (ABOM) extends a BOM by attaching risk attributes, test options, detection probabilities, costs, and schedule impacts, enabling comparative analysis rather than a single "optimal" answer.

- **⚠️** Hardware-test applicability assumption likely inflates detection performance.
  - Issue: The model assumes all hardware tests can detect vulnerabilities for all hardware ("over-estimate hardware vulnerability detection rate"), which could systematically bias decisions.
  - Recommendation (actionable): Add an optional test-to-component applicability matrix (even coarse, e.g., by component class) and demonstrate the impact on residual risk when applicability is restricted.
- **⚠️** Decision-support contribution needs a clearer "user workflow" and example decision question(s).
  - Issue: The report says the tool supports program offices, but the application section should more directly answer: "What decision does the PM/T&E lead make differently tomorrow because of this tool?"
  - Recommendation (actionable): Add 1-2 concrete use cases with an explicit decision statement, such as:
    - "Given a 30% schedule cut, which tests do we drop and what is the estimated leak-by increase (with CI)?"
    - "Which components should receive high-fidelity testing vs. screening under capacity constraints?" Include a short vignette and show the exact model outputs used in the decision.
- **⚠️** Quantification of "risk" should be better aligned to accepted risk constructs.
  - Issue: Current "impact-weighted risk exposure" is understandable, but it would benefit from clearer interpretation and units (e.g., expected impact points, expected compromised items, or an expected loss proxy).
  - Recommendation (actionable): Define the risk metric formally (equations and units), justify why it is decision-relevant, and add a short interpretation guide for sponsors.

### Contribution to Naval/DoD Mission

- **✅** Operational relevance is clear: the capstone addresses cybersecurity assurance planning and supply chain risk for microelectronics in DoD systems, explicitly tied to acquisition realities (limited budget/schedule/test capacity) and risk acceptance decisions.
- **✅** Stakeholder touchpoints are present (acknowledged AFRL sponsorship and Crane/DASN(RDT&E) inputs), which supports mission alignment.
- **⚠️** Sponsor alignment is not fully integrated into the technical narrative.



## Automatic Graders

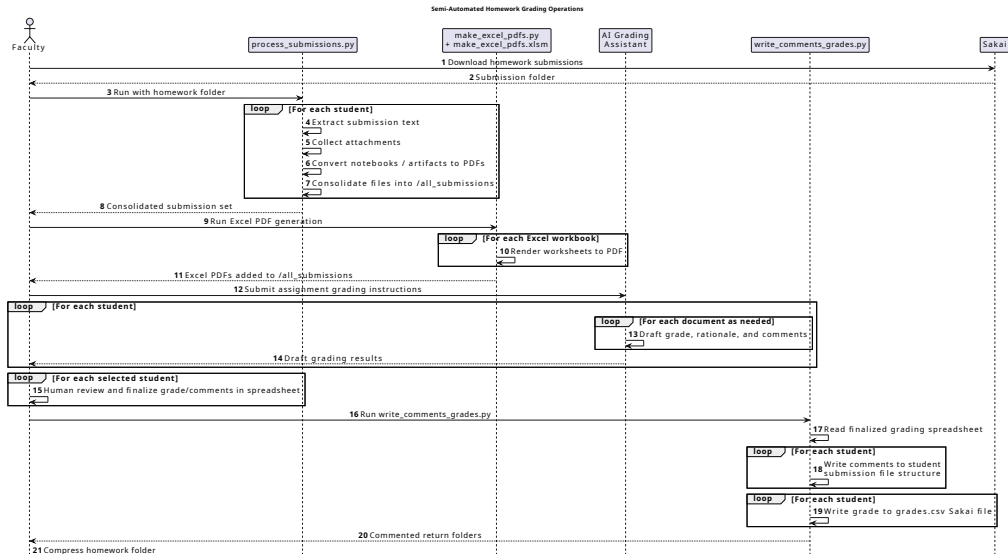
- Course-level grading assistants applied to quantitative SE homework assignments including Custom GPTs, portable chat instructions, API programs for batches, and local LLMs (in-process).
- Apply assignment-specific rubrics, identifies missing reasoning, and drafts formative feedback.
- Initial experiments compare AI-generated feedback with instructor grading and student revision outcomes.
- Outputs include a review narrative, rubric scores, common-error summaries, and machine-readable records for uploading of grades and comments.

## Grader Friction

- Multiple batches forced by ChatGPT limits on file uploads varying from 10-20 or sometimes 0.
- Variable performance by ChatGPT.
- Initial output for spreadsheet XML format took exceedingly long per batch.
- Fragility of the the XML spreadsheet output.
- Many regressive cycles while fine tuning instructions and fixing broken data structures.
- Combined friction forced development of API program to process an entire class at once.

- User: *This XML spreadsheet output takes way too long to print out and is fragile. Some batches take 15 minutes and sometimes your spreadsheets aren't readable. Can you suggest an alternate format that we can then convert to Excel?*
- ChatGPT: *Yes. I suggest a Python list of lists for these reasons ...*
- Outputs are now instantaneous and never break.

# Grading Workflow Sequence



## Automated Grading Processing Constraints

- Student submissions are accepted in any format, so grading starts with artifact normalization rather than AI review.
- Most submitted artifacts are converted to PDF for reliable AI parsing and consistent visual review. Colab URLs, `.ipynb` notebooks, Excel files, and other attachments are converted into reviewable PDF artifacts.
- Text-native files such as `.txt` and `.md` can be used directly without conversion.
- Excel requires special handling: embedded graphics, charts, text boxes, formatting, and multi-tab layouts. An Excel macro is used to render each workbook faithfully.
- Sakai dictates a nested per-student folder structure with submission text, attachments, timestamps, and other files, requiring automated traversal and consolidation.

# Example AI Grader Output

| Student Name        | Submission Text          | #5 Effort | #5 Schedule | #5 Cost     | #5 Avg Staffing | ... | Total Points | Comments                                                                                                                                     |
|---------------------|--------------------------|-----------|-------------|-------------|-----------------|-----|--------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ANSWER KEY</b>   | —                        | 145.1 PM  | 7.8 Months  | \$1,741,761 | 18.6 People     | ... | —            | —                                                                                                                                            |
| Doe, John           | —                        | 225 PM    | 9 Months    | \$2,700,261 | 25 People       | ... | 19.0         | John Doe (19/22)<br>5. 4/5 Missed requirements...<br>6. 4/4 Correct annual savings...<br>7. 2/3 Used a size weight for cost...               |
| Assaley, Emily      | —                        | 163.5 PM  | 8.1 Months  | \$1,962,429 |                 | ... | 18.0         | Emily Assaley (18/22)<br>5. 4/5 Missed requirements...<br>6. 4/4 Correct annual savings...<br>7. 3/3 Correct reuse model...                  |
| Astrup, Dylan       | See attached spreadsheet | 163.5 PM  | 8.1 Months  | \$1,962,429 | 20.2 People     | ... | 19.0         | Dylan Astrup (19/22)<br>5. 4/5 Missed requirements...<br>6. 4/4 Correct annual savings...<br>7. 2/3 Used a size weight for cost...           |
| Bernstein, Nicholas | —                        | 145.1 PM  | 7.8 Months  | \$1,741,761 | —               | ... | 20.0         | Nicholas Bernstein (20/22)<br>5. 5/5 Correct revised COSYSMO...<br>6. 4/4 Correct annual savings...<br>7. 2/3 Used a size weight for cost... |

## Automated Grading Experience Summary

- Applied to assignments in *Modeling and Simulation for Acquisition* (7) and *Engineering Economics and Cost Estimation* (5).
- Graded ~340 student assignments for ~6100 student points.
- **Zero** student complaints or comments about using AI for grading.
- **One** instance of a student questioning why a problem was marked wrong which was then fixed by giving back a point.
- First time through for each required multiple iterations to fine tune assignment-specific instructions.
- Extensive human review of AI outputs for each assignment, always checking points off and spot checking others.
- Total time for each took more than grading manually, but savings are expected the next time.
- ChatGPT downtime forced creation of portable instructions for other LLMs (some assignment had combined LLM outputs with

## Automated Grader: Early Operating Experience

- Each assignment has required substantial instruction tailoring because problem types, grading criteria, artifacts, and expected solution forms vary widely.
- Revamping the grader instructions for every assignment has been one of the largest setup costs.
- Going forward, the grader is being reorganized around reusable instruction components with transclusion, so common grading policies, output formats, and review rules do not have to be rewritten.
- AI grading has always required substantial human review in early operations to verify, correct, and calibrate scores and comments.
- The workflow is therefore semi-automated: AI drafts grades and feedback, while the instructor remains responsible for final grades.
- Across several thousand graded problems, no students have



# Summaries

## Governance and Sustainment

- Instructions and knowledge bases are treated as controlled engineering artifacts, not informal prompts.
- HEATHER 2 updates are managed through a formal configuration management board.
- Assistant baselines include intended users, authoritative sources, prohibited behaviors, output formats, and test cases.
- Sustainment uses a DevOps continuous process: collect issues, revise sources, test behavior, approve updates, and archive baselines.
- This reduces risk as staff change, procedures evolve, and model capabilities improve over time.

## Integrated Lessons

- AI academic value comes from combining institutional grounding, constrained behavior, and human review.
- SOPs, rubrics, and assessment instruments become executable knowledge sources when paired with careful instructions.
- Structured outputs connect individual assistance to program assessment and continuous improvement.
- Bootstrapping creates leverage: one well-governed assistant can help produce the next one.
- Trust depends on disclosure, monitoring, configuration control, and clear limits on AI authority.

## Conclusions

- HEATHER 2, the Capstone/Thesis Reviewer, Automatic Grader, and AI Bootstrapper form a coherent SE academic AI ecosystem.
- The common pattern is governed decision support for consequential academic processes.
- Current activities are strengthening configuration control, empirical evaluation, rubric calibration, student disclosure, and secure deployment planning.
- The integrated effort improves process continuity, assessment evidence, instructional feedback, and faculty/staff efficiency.
- Formal configuration management board will be essential for knowledge-base and instruction updates.
- The SE Department is demonstrating an early adopter path that can later be validated and adapted across campus.